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MODIFIED MAXIMUM LIKELIHOOD AND MODIFIED MOMENT ESTIMATORS
FOR THE THREE-PARAMETER WEIBULL DISTRIBUTION

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ABSTRACT

This article is concerned with modifications of both maximum likelihood and moment estimators for parameters of the three-parameter Weibull distribution. Modifications presented here are basically the same as those previously proposed by the authors (1980, 1981, 1982) in connection with the lognormal and the gamma distributions. Computer programs were prepared for the practical application of these estimators and an illustrative example is included. Results of a simulation study provide insight into the sampling behavior of the new estimators and include comparisons with the traditional moment and maximum likelihood estimators. For some combinations of parameter values, some of the modified estimators considered here enjoy advantages over both moment and maximum likelihood estimators with respect to bias, variance, and/or ease of calculation.

1. INTRODUCTION

The Weibull distribution is widely employed as a model for distributions of life spans and reaction times. From a computational point of view, it is particularly appealing since its

cumulative distribution function can be expressed explicitly as a simple function of the random variable. Both two and three parameter Weibull distributions have been studied extensively by numerous writers including Cohen (1965, 1973, 1975), Dubey (1966), Harter and Moore (1965, 1967), Bain (1978), Mann (1968), Mann and Fertig (1975a, 1975b), Lemon (1974), Wingo (1973), Ringer and Sprinkle (1972), Rockette (1974), Zanakis (1977, 1979a, 1979b), Zanakis and Mann (1981), Wycoff, Bain, and Englehardt (1980), Königer (1981), and various others.

Moment estimators (ME) and maximum likelihood estimators (MLE) have been considered by most of these previous writers to be the traditional or standard estimators for parameters of this distribution. It is well known, however, that for some combinations of parameter values, maximum likelihood estimation in the three-parameter Weibull does not satisfy the usual regularity conditions and consequently can give rise to estimates not having the usual asymptotic properties and even to inconsistent estimates. For some samples, maximum likelihood estimates simply fail to exist. When any of these conditions prevail, the modified estimators presented in this article provide alternatives which often enjoy advantages over both moment and maximum likelihood estimators. In most instances modified estimates are easier to calculate and frequently exhibit smaller biases and variances than the maximum likelihood estimates. The estimators proposed here for the Weibull parameters are similar to those previously proposed by the authors (1980, 1981, 1982) for parameters of the lognormal and the gamma distributions. Consideration here is limited to complete samples. Estimation based on censored samples was previously studied by Cohen (1965 and 1975).

Let X designate a random variable that is Weibull (γ, δ, β) and the probability density function can be written as

$$f(x; \gamma, \delta, \beta) = \frac{\delta}{\beta} (x - \gamma)^{\delta-1} \exp - [(x - \gamma)/\beta]^{\delta}, \quad (1.1)$$

for $\gamma < x < \infty$, $\delta > 0$, $\beta > 0$ (0, otherwise). The corresponding

cumulative distribution function is

$$F(x; \gamma, \delta, \beta) = 1 - \exp\{-[(x-\gamma)/\beta]^\delta\}. \quad (1.2)$$

In the notation employed here, γ is the location or threshold parameter, δ is the shape parameter, and β is the scale parameter. The mean μ_x , variance σ_x^2 , median Me_x and third standard moment $\alpha_{3;x}$ are

$$\mu_x = \gamma + \beta \Gamma_1, \quad \sigma_x^2 = \beta^2 [\Gamma_2 - \Gamma_1^2], \quad (1.3)$$

$$Me_x = \gamma + \beta (\ln 2)^{1/\delta}, \quad \alpha_{3;x} = \frac{\Gamma_3 - 3\Gamma_2\Gamma_1 + 2\Gamma_1^3}{[\Gamma_2 - \Gamma_1^2]^{3/2}},$$

where

$$\Gamma_k = \Gamma(1 + k/\delta), \quad (1.4)$$

and where $\Gamma(z)$ is the gamma function

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt. \quad (1.5)$$

In certain instances it is expedient to substitute θ for the scale parameter β , where

$$\theta = \beta^\delta \text{ and } \beta = \theta^{1/\delta}. \quad (1.6)$$

Although the Weibull distribution is usually perceived to be positively skewed, it becomes negatively skewed when $\delta > \delta_0 = 3.6023494257197$ (Cohen, 1973). It is, of course, positively skewed when $\delta < \delta_0$ and there is no skewness ($\alpha_{3;x}=0$) when $\delta = \delta_0$. In applications concerning life spans and reaction times, values of δ in excess of 3.22 ($\alpha_{3;x} < 0.10$) seldom occur. However, in other applications larger values of δ are sometimes encountered. Excellent expository accounts of this distribution and its applications have been given by Johnson and Kotz (1970) and by Mann, Schafer and Singpurwalla (1974).

2. MAXIMUM LIKELIHOOD ESTIMATORS (MLE)

The likelihood function of an ordered random sample $\{x_i\}$, $i = 1, 2, \dots, n$ from a distribution with p.d.f. (1.1) is

$$L(x_1, x_2, \dots, x_n; \gamma, \delta, \beta) = \left(\frac{\delta}{\beta}\right)^n \prod_{i=1}^n (x_i - \gamma)^{\delta-1} \exp \left\{ -\sum_{i=1}^n \left(\frac{x_i - \gamma}{\beta} \right)^\delta \right\}. \quad (2.1)$$

When $\delta < 1$, the distribution is reverse J-shaped and the likelihood function becomes infinite as $\gamma \rightarrow x_1$, the smallest sample observation. Accordingly in this situation, the maximum likelihood estimate of γ would be x_1 , but estimates of β and δ would not exist.

When $\delta = 1$, the two-parameter exponential distribution emerges from (1.1) as a special case and as shown by Cohen and Helm (1973), estimators that are both best linear unbiased (BLUE) and minimum variance unbiased (MVUE) are

$$\hat{\beta} = n(\bar{x} - x_1)/(n-1) \text{ and } \hat{\gamma} = (nx_1 - \bar{x})/(n-1), \quad (2.2)$$

where $\bar{x} = \sum_{i=1}^n x_i/n$.

When $\delta > 1$, the Weibull distribution is bell-shaped and maximum likelihood estimators can be found by simultaneously solving the system of equations obtained by equating to zero the partial derivatives of the log likelihood function with respect to the parameters. Computational problems are likely to be encountered when δ is near to 1, even though δ actually exceeds 1. Furthermore, the usual asymptotic properties of MLE do not hold unless $\delta > 2$. As a practical matter, it is not considered advisable to employ the MLE for parameters of the three-parameter Weibull distribution unless there is reason to expect that $\delta > 2.2$. Of course the exact variances given in Section 6 are applicable in the special case when δ is known to equal 1.

On taking logarithms of (2.1), differentiating and equating to zero, we obtain

$$\begin{aligned}
\frac{\partial \ln L}{\partial \gamma} &= \frac{\delta}{\theta} \sum_1^n (x_i - \gamma)^{\delta-1} - (\delta-1) \sum_1^n (x_i - \gamma)^{\delta-2} = 0, \\
\frac{\partial \ln L}{\partial \delta} &= \frac{n}{\delta} + \sum_1^n \ln(x_i - \gamma) - \frac{1}{\theta} \sum_1^n (\gamma_i - \gamma)^{\delta} \ln(x_i - \gamma) = 0, \\
\frac{\partial \ln L}{\partial \theta} &= -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_1^n (x_i - \gamma)^{\delta} = 0.
\end{aligned} \tag{2.3}$$

When δ is sufficiently large as indicated previously, the MLE, $\hat{\gamma}$, $\hat{\delta}$, $\hat{\theta}$ follow as the simultaneous solution of (2.3). If an estimate of β is desired, it follows from (1.6) as $\hat{\beta} = \hat{\theta}^{1/\hat{\delta}}$.

The three equations of (2.3) do not yield explicit solutions for the estimates. However, as shown by Cohen (1965), θ can be eliminated from the last two equations of (2.3) to give

$$\left[\frac{\sum_1^n (x_i - \gamma)^{\delta} \ln(x_i - \gamma)}{\sum_1^n (x_i - \gamma)^{\delta}} - \frac{1}{\delta} \right] - \frac{1}{n} \sum_1^n \ln(x_i - \gamma) = 0. \tag{2.4}$$

When γ is known, this equation can be solved iteratively for $\hat{\delta}$. The chart of δ as a function of the sample coefficient of variation given by Cohen (1965) may be useful in finding a first approximation to $\hat{\delta}$ for use in solving (2.4). An alternate first approximation might be found by interpolating in Table I with $\alpha_3 = a_3$ where $a_3 = \sum_1^n (x_i - \bar{x})^3 / ns^3$, and $s^2 = \sum_1^n (x_i - \bar{x})^2 / (n-1)$. Subsequently $\hat{\theta}$ follows from the third equation of (2.3) as

$$\hat{\theta} = \frac{1}{n} \sum_1^n (x_i - \gamma)^{\hat{\delta}}. \tag{2.5}$$

When γ is unknown and must therefore be estimated from the sample data, we select a first approximation $\gamma_1 (< x_1)$ and employ a "trial and error" procedure. (Various other iterative procedures might have been used.) With γ_1 fixed, (2.4) is solved for δ_1 and θ_1 follows from (2.5). We then calculate $(\partial \ln L / \partial \gamma)_1$ by substituting γ_1 , δ_1 and θ_1 into the first equation of (2.3). If $(\partial \ln L / \partial \gamma)_1 = 0$, then $\hat{\gamma} = \gamma_1$, $\hat{\delta} = \delta_1$, $\hat{\theta} = \theta_1$, and our task is complete. Otherwise, we repeat the cycle of computations with a new approximation γ_2 and continue until we find a pair of values (γ_i, γ_j) such that $|\gamma_i - \gamma_j|$ is sufficiently small and such

that $(\partial \ln L / \partial \gamma)_i \leq 0 \leq (\partial \ln L / \partial \gamma)_j$. We then interpolate linearly for the required estimates.

3. MODIFIED MAXIMUM LIKELIHOOD ESTIMATORS (MMLE)

The modified estimators MMLE considered here were originally offered as alternates for use when the MLE were likely to be unsatisfactory; i.e. when $\delta < 2.0$ (or perhaps 2.2). However some of them appear to offer advantages with respect to ease of computation, bias and/or variance for all values of δ . The various modified estimators under consideration were obtained by replacing the equation $\partial \ln L / \partial \gamma = 0$, (i.e. the first equation of (2.3)) with alternate functional relationships as was done by the authors (1980, 1981, 1982) in connection with parameter estimation in the lognormal and the gamma distributions.

3.1. Concerning MMLE-I

In this case, as estimating equations, we employ $\partial \ln L / \partial \delta = 0$, $\partial \ln L / \partial \theta = 0$ and $E[F(X_r)] = F(x_r)$, where $E(\cdot)$ is the usual expectation symbol, X_r is the r th order statistic in a random sample of size n , and $F(X_r)$ is the associated value of the cumulative distribution. Since

$$E[F(X_r)] = r/(n+1), \quad (3.1)$$

this equation becomes

$$1 - \exp[-(x_r - \gamma)^\delta / \theta] = r/(n+1), \quad (3.2)$$

a result which reduces to

$$-\ln[(n+1-r)/(n+1)] = (x_r - \gamma)^\delta / \theta. \quad (3.3)$$

Although r might assume any value from 1 to n , at this time we restrict further consideration to $r = 1$. The first order statistic contains more information about γ than any of the other order statistics and often more information than all the other order statistics combined. Of course, when outliers are present or are suspected of being present, as could happen when data are contaminated, values of $r > 1$ might be justified.

With $r = 1$, (3.3) becomes

$$-\ln[n/(n+1)] = (x_1 - \gamma)^\delta / \theta. \quad (3.4)$$

The estimating equations to be solved in this case are accordingly (2.4), (2.5) and (3.4). The computational procedure is essentially the same as that employed for the MLE. We choose trial values of $\gamma(<x_1)$ and proceed until two values γ_i and γ_j are found such that $|\gamma_i - \gamma_j|$ is sufficiently small and such that

$$(x_1 - \gamma_i)^{\delta_i / \theta_i} \geq -\ln[n/(n+1)] \geq (x_1 - \gamma_j)^{\delta_j / \theta_j}.$$

We then interpolate linearly for the required estimates.

It is to be noted that the MMLE-I and other modified estimators are applicable regardless of whether δ is equal to, less than, or greater than 1. In all cases, however, it is necessary that $\hat{\gamma} < x_1$.

3.2. Concerning MMLE-II

In this case the first equation of (2.3) is replaced by $E(X_1) = x_1$. It is well known [c.f., for example, Cohen (1975)] that

$$E(X_1) = \gamma + [\beta/n^{1/\delta}] \Gamma_1. \quad (3.5)$$

Accordingly the estimating equations are (2.4), (2.5) and

$$\gamma + [\beta/n^{1/\delta}] \Gamma_1 = x_1. \quad (3.6)$$

The required estimates follow by linear interpolation after we find two values γ_i and γ_j such that $|\gamma_i - \gamma_j|$ is sufficiently small and such that

$$\gamma_i + [\beta_i/n^{1/\delta_i}] \Gamma_{1(i)} \geq x_1 \geq \gamma_j + [\beta_j/n^{1/\delta_j}] \Gamma_{1(j)}.$$

3.3. Concerning MMLE-III

In this case, the first equation of (2.3) is replaced by $E(X) = \bar{x}$, and we thus have

$$\gamma + \beta \Gamma_1 = \bar{x}. \quad (3.7)$$

The estimating equations to be solved, accordingly consist of (2.4), (2.5) and (3.7). As in the previous modifications, we interpolate between values γ_i and γ_j where $|\gamma_i - \gamma_j|$ is sufficiently small and such that

$$\gamma_i + \beta_i \Gamma_1(i) \geq \bar{x} \geq \gamma_j + \beta_j \Gamma_1(j).$$

3.4. Concerning MMLE-IV

This time, the first equation of (2.1) is replaced by $V(X) = s^2$, where s^2 is the sample variance and $V(X) = \sigma_x^2$ is the population variance. Accordingly, we have

$$\beta^2 [\Gamma_2 - \Gamma_1^2] = s^2. \quad (3.8)$$

Thus our three estimating equations consist of (2.4), (2.5) and (3.8). The required estimates follow by linear interpolation between γ_i and γ_j where $|\gamma_i - \gamma_j|$ is sufficiently small and such that

$$\beta_i^2 [\Gamma_2(i) - \Gamma_1^2(i)] \geq s^2 \geq \beta_j^2 [\Gamma_2(j) - \Gamma_1^2(j)].$$

3.5. Concerning MMLE-V

For this case, the first equation of (2.1) is replaced by

$$Me_x = x_{me},$$

where Me_x is the population median and x_{me} is the sample median. Thus, we have

$$\gamma + \beta(\ln 2)^{1/\delta} = x_{me} \quad (3.9)$$

The estimating equations accordingly consist of (2.4), (2.5) and (3.9). The required estimates follow by linear interpolation between γ_i and γ_j such that

$$\gamma_i + \beta_i (\ln 2)^{1/\delta_i} \geq x_{me} \geq \gamma_j + \beta_j (\ln 2)^{1/\delta_j},$$

where $|\gamma_i - \gamma_j|$ is sufficiently small.

4. MOMENT (ME) AND MODIFIED MOMENT ESTIMATORS (MME)

As is well known, the moment estimators are obtained by equating the first three population moments to corresponding sample moments. It accordingly follows that

$$\gamma + \beta\Gamma_1 = \bar{x}, \quad \beta^2[\Gamma_2 - \Gamma_1^2] = s^2, \quad (4.1)$$

$$\frac{\Gamma_3 - 3\Gamma_2\Gamma_1 + 2\Gamma_1^3}{[\Gamma_2 - \Gamma_1^2]^{3/2}} = a_3.$$

The moment estimate δ^* is obtained by solving the third equation of (4.1). This can be accomplished by employing standard iterative techniques. A first approximation to the required solution can be obtained by interpolation in Table I. With δ^* thus calculated, β^* follows from the second equation of (4.1) as

TABLE I
 δ As a Function of α_3

α_3	δ	α_3	δ	α_3	δ
		.30	2.63756	1.70	1.11461
		.40	2.41032	1.75	1.09326
-1.00	40.74307	.50	2.21560	1.80	1.07289
-.90	22.92716	.60	2.04757	1.90	1.03483
-.80	15.62697	.70	1.90169	2.00	1.00000
-.70	11.66381	.80	1.77431	2.10	.96805
-.60	9.18473	.90	1.66251	2.20	.93864
-.50	7.49353	1.00	1.56391	2.30	.91152
-.40	6.27095	1.10	1.47657	2.40	.88643
-.30	5.34979	1.20	1.39885	2.50	.86317
-.20	4.63397	1.25	1.36318	2.60	.84156
-.10	4.06431	1.30	1.32943	2.70	.82143
0	3.60235	1.40	1.26717	2.80	.80265
.10	3.22197	1.50	1.21112	2.90	.78508
.20	2.90481	1.60	1.16050	3.00	.76862

$$\beta^* = s / \sqrt{\Gamma_2(\delta^*) - \Gamma_1^2(\delta^*)}, \quad (4.2)$$

and γ^* follows from the first equation as

$$\gamma^* = \bar{x} - \beta^* \Gamma_1(\delta^*). \quad (4.3)$$

4.1. Concerning MME-I

In this modification of the moment estimators, the equation involving the third sample moment, (i.e. the third equation of (4.1)) is replaced by

$$E[F(X_r)] = F(x_r).$$

As in the case of MMLE-I, we restrict our concern to $r=1$, and the third estimating equation becomes

$$-\ln[n/(n+1)] = [(x_1 - \gamma)/\beta]^\delta. \quad (4.4)$$

On eliminating β and γ from the three estimating equations, we obtain

$$\frac{s^2}{(\bar{x} - x_1)^2} = \frac{\Gamma_2 - \Gamma_1^2}{[\Gamma_1 - \{-\ln[n/(n+1)]\}^{1/\delta}]^2}. \quad (4.5)$$

This equation can be solved for δ^* by using standard iterative procedures or even by using a simple "trial and error" technique. A first approximation to δ^* might be obtained by interpolation in Table I with a_3 set equal to α_3 . With δ^* determined from (4.5), β^* and γ^* follow from (4.2) and (4.3). This estimator was first proposed by F. Russell Helm (1970) in his doctoral dissertation. As an aid in the solution of (4.5) for δ^* , he constructed tables of δ as a function of the arguments n and $s^2/(\bar{x} - x_1)^2$ for selected values of the arguments.

4.2. Concerning MME-II

In this case, the third equation of (4.1) is replaced by

$$E(X_1) = x_1,$$

which reduces to

$$\gamma + \{\beta/n^{1/\delta}\}\Gamma_1 = x_1 \quad (4.6)$$

Elimination of β and γ from the three applicable estimating equations yields

$$\frac{s^2}{(\bar{x} - x_1)^2} = \frac{\Gamma_2 - \Gamma_1^2}{[(1 - n^{-1/\delta})\Gamma_1]^2} \quad (4.7)$$

This equation can be solved for δ^* by using techniques similar to those employed in the solution of (4.5). With δ^* thus determined, β^* and γ^* again follow from (4.2) and (4.3).

4.3. Concerning MME-III

In this modification, the third equation of (4.1) is replaced by

$$Me_x = x_{me} ,$$

and thus we have

$$\gamma + \beta(\ln 2)^{1/\delta} = x_{me} \quad (4.8)$$

Elimination of β and γ from the three applicable estimating equation yields

$$\frac{s^2}{(\bar{x} - x_{me})^2} = \frac{\Gamma_2 - \Gamma_1^2}{[\Gamma_1 - (\ln 2)^{1/\delta}]^2} \quad (4.9)$$

Equation (4.9) can be solved for δ^* by using the same techniques as those employed in solving (4.5) and (4.7). Again β^* and γ^* follow from (4.2) and (4.3).

5. SOME SPECIAL CASES

When one of the three parameters is known, estimation of the two remaining parameters is correspondingly simplified.

5.1. Threshold Parameter γ Is Known

When γ is known, the p.d.f. of (1.1) reduces to the usual two-parameter distribution. The MLE for this case was given by

Cohen (1965). It is necessary only that (2.4) be solved for $\hat{\delta}$ and that $\hat{\theta}$ be calculated from (2.5).

The various modified maximum likelihood estimators employ $\partial \ln L / \partial \theta = 0$ plus either (3.4), (3.6), (3.7), (3.8) or (3.9). When θ is eliminated between the two applicable estimating equations, we obtain results which follow.

MMLE-I

$$\frac{n(x_1 - \gamma)^\delta}{-\ln[n/(n+1)]} = \sum_1^n (x_i - \gamma)^\delta. \quad (5.1)$$

MMLE-II

$$n^2 \left(\frac{x_1 - \gamma}{\Gamma_1} \right)^\delta = \sum_1^n (x_i - \gamma)^\delta. \quad (5.2)$$

MMLE-III

$$n \left(\frac{\bar{x} - \gamma}{\Gamma_1} \right)^\delta = \sum_1^n (x_i - \gamma)^\delta. \quad (5.3)$$

MMLE-IV

$$n \left(\frac{s^2}{\Gamma_2 - \Gamma_1^2} \right)^{\delta/2} = \sum_1^n (x_i - \gamma)^\delta. \quad (5.4)$$

MMLE-V

$$\frac{n(x_{me} - \gamma)^\delta}{\ln 2} = \sum_1^n (x_i - \gamma)^\delta. \quad (5.5)$$

In each of these cases with γ known, we solve for $\hat{\delta}$. Subsequently $\hat{\theta}$ follows from (2.5). Corresponding simplifications follow for the moment and modified moment estimators when γ is known.

5.2. Shape Parameter δ Is Known

The estimating equations in this instance are identically the same as those given in Sub-Section 5.1 for use when γ is known. This time, however, with δ known, we would solve (5.1),

(5.2), (5.3), (5.4), or (5.5) for $\hat{\gamma}$. The estimate, $\hat{\theta}$, then follows from (2.5) as before.

6. ASYMPTOTIC VARIANCES AND COVARIANCES

Asymptotic variances and covariances which apply in the case of maximum likelihood estimators when $\delta > 2$, can be obtained by inverting the Fisher information matrix with elements that are negatives of expected values of second partials of the log likelihood function. Accordingly, in this case, the variance-covariance matrix, $V(\hat{\gamma}, \hat{\delta}, \hat{\theta})$, becomes

$$V = \begin{vmatrix} \frac{n(\delta-1)}{\delta^2} [r(1-\frac{2}{\delta}) + \delta r(2-\frac{2}{\delta})] & \frac{n}{\delta} [r(1-\frac{1}{\delta}) - r(2-\frac{1}{\delta})(1+\psi(2-\frac{1}{\delta}))] & \frac{n\delta^2}{\delta^2} r(2-\frac{1}{\delta}) \\ \frac{n}{\delta} [r(1-\frac{1}{\delta}) - r(2-\frac{1}{\delta})(1+\psi(2-\frac{1}{\delta}))] & \frac{n}{\delta^2} [\psi'(1) + (\psi(2))^2] & -\frac{n}{\delta} \psi(2) \\ \frac{n\delta^2}{\delta^2} r(2-\frac{1}{\delta}) & -\frac{n}{\delta} \psi(2) & \frac{n\delta^2}{\delta^2} \end{vmatrix}^{-1} \quad (6.1)$$

where $\Psi(\cdot)$ is the digamma function, $\Psi(z) = d[\ln\Gamma(z)]/dz = \Gamma'(z)/\Gamma(z)$ and $\Psi'(z)$ is the trigamma function. Tables of these functions for values of the argument ranging from 1.000 to 2.000 at intervals of .005 are given by Abramowitz and Stegun (1964).

For the estimates $\hat{\gamma}$, $\hat{\delta}$, and $\hat{\theta}$, the matrix $V(\hat{\gamma}, \hat{\delta}, \hat{\theta})$ becomes

$$V = \begin{vmatrix} \frac{n(\delta-1)}{\delta^{2/\delta}} [r(1-\frac{2}{\delta}) + \delta r(2-\frac{2}{\delta})] & \frac{n}{\delta^{1/\delta}} [r(1-\frac{1}{\delta}) - r(2-\frac{1}{\delta})(1+\psi(2-\frac{1}{\delta})+\epsilon_{no})] & \frac{n\delta}{\delta^{1+1/\delta}} r(2-\frac{1}{\delta}) \\ \frac{n}{\delta^{1/\delta}} [r(1-\frac{1}{\delta}) - r(2-\frac{1}{\delta})(1+\psi(2-\frac{1}{\delta})+\epsilon_{no})] & \frac{n}{\delta^2} [\psi'(1) + (\psi(2) + \epsilon_{no})^2] & -\frac{n}{\delta^2} [\psi(2) + \epsilon_{no}] \\ \frac{n\delta}{\delta^{1+1/\delta}} r(2-\frac{1}{\delta}) & -\frac{n}{\delta^2} [\psi(2) + \epsilon_{no}] & \frac{n}{\delta^2} \end{vmatrix}^{-1} \quad (6.2)$$

Although (6.1) and (6.2) are valid if $\delta > 2$, computational difficulties might be experienced unless δ is approximately 2.2 or greater.

In the two-parameter special case where γ is known, the first row and the first column of (6.1) and also of (6.2) are eliminated. On completing the inversion of the resulting 2x2 matrices, we obtain

$$\begin{aligned}
V(\hat{\delta}) &= \frac{\delta^2}{n \Psi'(1)}, \\
V(\hat{\beta}) &= \frac{\beta^2}{n \delta^2} \left[1 + \frac{(\Psi(2))^2}{\Psi'(1)} \right], \\
\text{Cov}(\hat{\delta}, \hat{\beta}) &= \frac{\beta}{n} \left[\frac{\Psi(2)}{\Psi'(1)} \right], \\
V(\hat{\theta}) &= \frac{\theta^2}{n} \left[1 + \frac{(\Psi(2) + \ln \theta)^2}{\Psi'(1)} \right], \\
\text{Cov}(\hat{\delta}, \hat{\theta}) &= \frac{\delta \theta}{n} \left[\frac{\Psi(2) + \ln \theta}{\Psi'(1)} \right].
\end{aligned} \tag{6.3}$$

Correlation coefficients follow as

$$\begin{aligned}
r_{\hat{\delta}, \hat{\beta}} &= \text{Cov}(\hat{\delta}, \hat{\beta}) / \sqrt{V(\hat{\delta}) V(\hat{\beta})} = \frac{\Psi(2)}{\sqrt{\Psi'(1) + (\Psi(2))^2}}, \\
r_{\hat{\delta}, \hat{\theta}} &= \frac{\Psi(2) + \ln \theta}{\sqrt{\Psi'(1) + (\Psi(2) + \ln \theta)^2}}.
\end{aligned} \tag{6.4}$$

On substituting the values $\Psi(2) = 0.4227843351$ and $\Psi'(1) = 1.6449340668$ (Abramowitz and Stegun, 1964), (6.3) and (6.4) become

$$\begin{aligned}
V(\hat{\delta}) &= 0.607927 \delta^2 / n, \\
V(\hat{\beta}) &= 1.108665 \beta^2 / \delta^2 n, \\
\text{Cov}(\hat{\delta}, \hat{\beta}) &= 0.257022 \beta / n, \\
V(\hat{\theta}) &= \frac{\theta^2}{n} \left[1 + \frac{(0.4227843 + \ln \theta)^2}{1.644934} \right], \\
\text{Cov}(\hat{\delta}, \hat{\theta}) &= \frac{\delta \theta}{n} \left[\frac{0.4227843 + \ln \theta}{1.644934} \right], \\
r_{\hat{\delta}, \hat{\beta}} &= 0.313072, \\
r_{\hat{\delta}, \hat{\theta}} &= \frac{0.422784 + \ln \theta}{\sqrt{1.644934 + (0.422784 + \ln \theta)^2}}.
\end{aligned} \tag{6.5}$$

Although the foregoing asymptotic results are in a strict sense applicable only for the maximum likelihood estimators,

simulation results reported in Section 8 indicate that they also provide reasonable approximations to variances and covariances of the modified estimators.

As previously noted, in the three-parameter distribution the asymptotic variances and covariances are valid only if $\delta > 2$. However, in the two-parameter distribution with γ known, they are valid for $\delta > 1$. In the special case of the two parameter exponential distribution (δ known to equal 1), the exact variances and covariance [c.f. Cohen and Helm (1973)] are

$$V(\hat{\beta}) = \frac{\beta^2}{n-1}, \quad V(\hat{\gamma}) = \frac{\beta^2}{n(n-1)}, \quad \text{Cov}(\hat{\beta}, \hat{\gamma}) = \frac{-\beta^2}{n(n-1)}. \quad (6.6)$$

Since the asymptotic variances and covariances are of limited applicability and since exact variances and covariance are available only when $\delta = 1$ (known), a Monte Carlo study has been included in order to assess the sampling behavior of the various modified estimators. Details are given in Section 8.

7. COMPUTATIONAL PROCEDURES

Most of the calculations required in practical applications of estimators presented here can be performed on a hand-held calculator by using tables of the gamma, digamma and trigamma functions such as those given by Abramowitz and Stegun (1964). Some of the programmable hand-held calculators can be programmed to calculate these functions and thereby eliminate dependency on tables. Although standard iterative procedures might be employed, simple "trial and error" techniques coupled with linear interpolation are adequate for solving the various estimating equations. However, in order to facilitate routine calculation of estimates, a FORTRAN software package similar to those previously developed by the authors (1980, 1981, 1982) for parameters of the lognormal and gamma distributions, was prepared. This package was used to produce the simulation summaries contained in Tables II and III. (Computer listings of this package may be obtained from the authors.)

The complete package consists of six subprograms which are grouped into three modules, a BASIC module, a MLE module and a

TABLE II
Simulation Summary - Mean Values of Parameter Estimates
Population Parameters: $\mu=0, \sigma=1.0000$

ESTIMATOR	100 Samples				200 Samples $n = 25$				10 Samples $n = 10$
	$\mu=4.5$ $\sigma=0.178$	$\mu=2.2156$ $\sigma=0.5$	$\mu=1.5639$ $\sigma=1.0$	$\mu=1.2111$ $\sigma=1.5$	$\mu=1.0000$ $\sigma=2.0$	$\mu=0.8632$ $\sigma=2.5$	$\mu=1.2111$ $\sigma=1.5$	$\mu=0.8632$ $\sigma=2.5$	$\mu=0.5$ $\sigma=6.6188$
ME	.0066 4.6775 .9909 0	-1.1595 2.3028 2.5461 0	-.5833 1.6504 1.8946 0	-.2756 1.3071 1.5288 0	-.0775 1.1083 1.3142 0	.0738 .9840 1.1869 0	-.2454 1.6880 1.3048 0	-.3796 1.3186 1.5392 0	.6092 .3079 2.6699 0
MME-I	-.2695 6.2672 1.2680 9	.1111 1.9265 .8726 0	.0467 1.4913 .9416 0	.0202 1.1995 .9733 0	.0094 1.0079 .9891 0	.0048 .8792 .9992 0	-.0582 1.4173 1.0716 0	-.0218 .9863 1.0564 0	-.0174 .6665 1.2336 0
MME-II	-.2488 6.0968 1.2479 10	-.0120 2.2333 1.0093 17	-.0131 1.6082 1.0152 0	-.0048 1.2350 1.0065 0	.0017 1.0165 .9998 0	.0032 .8805 1.0014 0	-.0627 1.4276 1.0761 0	-.0229 .9876 1.0580 0	-.0176 .6665 1.2337 0
MME-III	-.2835 6.2668 1.2817 4	-.0090 2.2501 1.0029 35	-.0975 1.7744 1.1141 20	-.1254 1.3894 1.1615 35	-.1563 1.1553 1.2156 39	-.1941 1.0026 1.2712 40	-.0215 1.2603 1.0282 0	-.1413 .9854 1.2469 11	-4.0209 .8956 .6072 9
MLE	.0048 4.5619 .9922 0	.0376 2.1232 .9527 6	.0237 1.5241 .9674 0	.0137 1.2111 .9837 17	N.A.	N.A.		N.A.	N.A.
MMLE-I	-.2459 5.3511 1.2420 0	-.0058 2.2281 .9992 22	-.0163 1.6338 1.0213 0	-.0067 1.2589 1.0113 4	-.0028 1.0277 1.0101 0	.0000 .8774 1.0113 0	-.0729 1.4826 1.0997 0	-.0239 .9969 1.0633 0	-.0087 .6110 1.1171 0
MMLE-IP	-.0726 4.5727 1.0666 0	.0063 2.2098 .9896 12	-.0065 1.6092 1.0087 0	-.0044 1.2533 1.0072 4	-.0024 1.0267 1.0094 0	-.0005 .8792 1.0129 0	-.0410 1.4211 1.0667 0	-.0222 .9949 1.0640 0	-.0122 .6161 1.1317 0
MMLE-III	-.3034 4.7337 1.3697 0	.0141 2.2054 .9808 3	.0020 1.5875 .9975 0	-.0020 1.2433 1.0024 4	-.3029 1.3851 1.3730 1	-.7486 1.5843 1.9603 1	-.0431 1.2849 1.0565 0	-1.4243 2.1571 2.7889 0	-.0711 .5569 1.1660 0
MMLE-IV	.0580 3.9063 .9288 0	.0097 2.2122 .9916 4	-.0004 1.5926 .9969 3	-.0124 1.2659 .9907 24	-.0155 1.0483 .9858 30	-.0132 .9134 1.0196 31	.0075 1.2259 .9921 0	-.0759 1.0090 1.1290 0	.0021 .5030 .9616 0
MMLE-V	-.7717 4.2240 1.7501 0	.0141 2.1789 .9806 40	-.0648 1.7243 1.0778 17	-.0518 1.3404 1.0750 29	-.0404 1.0949 1.0762 33	-.0351 .9546 1.0920 37	-.3426 1.7946 1.3710 0	-.5370 1.4440 1.6309 0	.0013 .5212 .9923 0

In each four-line entry, the top line pertains to μ , the second to σ , the third to σ and the bottom line is the number of samples which failed to yield estimates.

MOMENT module. The BASIC module consists of two programs, the first of which calculates the sample mean, standard deviation, third standard moment, first order statistic and median. The second program of this module evaluates the gamma function, $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$. The MLE module consists of subprograms which calculate maximum likelihood estimates, MLE, and the vari-

TABLE III
Simulation Summary - Standard Deviations of Parameter Estimates
Population Parameters: $\gamma=0$, $\rho=1.0000$

ESTIMATOR	100 Samples			n = 100			200 Samples n = 25		10 Samples n = 10
	$s=4.5$ $\alpha_3=-0.178$	$s=2.2156$ $\alpha_3=0.5$	$s=1.5639$ $\alpha_3=1.0$	$s=1.2111$ $\alpha_3=1.5$	$s=1.0000$ $\alpha_3=2.0$	$s=0.8632$ $\alpha_3=2.5$	$s=1.2111$ $\alpha_3=1.5$	$s=0.8632$ $\alpha_3=2.5$	$s=0.5$ $\alpha_3=6.6188$
ME	.2392 1.2765 .2528 0	.3373 .3847 .7400 0	.2132 .2616 .5620 0	.1668 .2099 .5119 0	.1546 .1816 .4983 0	.1641 .1636 .5002 0	.2360 .5393 .3629 0	.2547 .3884 .4935 0	.3712 .2747 2.0171 0
MMLE-I	.6134 3.7041 .6221 9	.0457 .1820 .0745 0	.0264 .1252 .0813 0	.0145 .1013 .0980 0	.0081 .0899 .1179 0	.0047 .0835 .1391 0	.1828 .6516 .2827 0	.0496 .2364 .2780 0	.0241 .1703 .6766 0
MMLE-II	.5690 3.2054 .5809 10	.0783 .2721 .1113 17	.0409 .1756 .0994 0	.0200 .1187 .1062 0	.0129 .0987 .1250 0	.0066 .0857 .1422 0	.2038 .7185 .2974 0	.0505 .2382 .2788 0	.0242 .1703 .6769 0
MMLE-III	.6853 3.6698 .6955 4	.1337 .3219 .1546 35	.1646 .3188 .2080 20	.1490 .2038 .2101 35	.1797 .1585 .2601 39	.2669 .1496 .3777 40	.1168 .1985 .2048 0	.1882 .1109 .2524 11	- - - 9
MLE	.2581 1.3969 .2719 0	.0682 .2609 .0988 6	.0331 .1603 .0927 0	.0157 .0910 .0975 17	N.A.	N.A.		N.A.	N.A.
MMLE-I	.4085 2.2788 .4117 0	.0696 .2487 .0969 22	.0422 .1832 .1021 0	.0196 .1099 .1078 4	.0102 .0898 .1215 0	.0088 .0738 .1118 0	.1485 .5307 .2707 0	.0531 .2504 .2800 0	.0104 .1376 .6163 0
MMLE-II	.3350 1.7057 .3360 0	.0705 .2642 .1006 12	.0382 .1717 .0968 0	.0188 .1074 .1067 4	.0098 .0884 .1218 0	.0088 .0789 .1118 0	.1009 .4010 .2355 0	.0442 .2271 .2751 0	.0137 .1486 .6316 0
MMLE-III	1.0078 4.0540 .9994 0	.0651 .2555 .0945 3	.0412 .1662 .0985 0	.0259 .1155 .1117 4	.6286 .6217 .7731 1	.8877 .8038 1.1156 1	.1949 .2152 .2470 0	.7063 .6098 .8612 0	.2330 .1745 .5723 0
MMLE-IV	1.0918 4.6026 1.1143 0	.0381 .1681 .0143 4	.0345 .1366 .0372 3	.0238 .0941 .0533 24	.0179 .0851 .0709 30	.0153 .0781 .0875 31	.0674 .2498 .2008 0	.4209 .2516 .6002 0	.0023 .0789 .5056 0
MMLE-V	1.7025 2.4052 1.6619 0	.1000 .2679 .1302 40	.1357 .2754 .1745 17	.0929 .1685 .1562 29	.0603 .1190 .1522 33	.0424 .1172 .1412 37	1.0176 1.5907 1.1158 0	1.4383 1.4303 1.6104 0	.0032 .1001 .5219 0
THEORETICAL VALUES	.2158* 1.1374 -.2235	.0368* .2068 -.0637	- .1219* .0674	- .0944* .0869	.0101** - .1005	.0101** - .1005	- .1889* .1739	.0408** - .2041	

In each four-line entry, the top line is α_1 , the second is α_2 , the third is α_3 and the bottom line is the number of samples which failed to yield estimates. *Asymptotic values calculated from (6.1). **Exact standard deviations calculated from (6.6) would apply if $s=1$, known. #Asymptotic values calculated from (6.5) would apply if γ were known.

ous modifications of the MLE. The MOMENT module consists of subprograms which calculate moment estimates, ME, and the various modifications of ME.

The BASIC module provides input for both the MLE and the MOMENT modules. The two latter modules are designed to solve applicable estimating equations iteratively for admissible estimates. For the maximum likelihood estimators and the modified maximum likelihood estimators, rules of admissibility require that $\hat{\gamma} < x_1$, and this restriction was incorporated into the MLE module. However, this restriction is not imposed on the moment and modified moment estimators. These estimators may therefore occasionally produce estimates of γ which are greater than x_1 . In order to reduce computer time, a further restriction is imposed on the MLE and its modifications which requires that estimates of γ fall within an interval extending from a lower limit that is ten standard deviations below the sample mean to an upper limit that is no greater than $x_1 - 10^{-4}$. When a maximum likelihood or modified maximum likelihood estimate of γ falls outside this interval, such an occurrence should be interpreted as a suggestion that the Weibull distribution might not be the most appropriate model for the data under consideration. This restriction does not apply to the moment or modified moment estimators.

Additional restrictions were imposed on estimates of δ . For the MLE, it is required that $\hat{\delta}$ fall in the interval (1.01, 15.0) while for all other estimators it is required that $\hat{\delta}$ fall in the interval (0.1, 15.0). The upper limit is somewhat arbitrary and can easily be adjusted to fit the needs of the user. For users whose data are fitted by distributions that are positively skewed, an upper limit of 3.6 might be considered since when $\hat{\delta} = 3.6$ the third standard moment is approximately 0. When restrictions on γ and/or $\hat{\delta}$ are violated, calculations are terminated, and an appropriate message is printed for the user. The necessity for restrictions such as those imposed here has also been noted by Zanakakis (1977).

A dichotomous search technique is employed in solving applicable estimating equations in both the MOMENT and MLE modules. For ME, and MME I-III, solutions to the third equation of (4.1), and solutions to equations (4.5), (4.7), and (4.9) are sought by first substituting $\hat{\delta}_i = .01$ into the appropriate equation and incrementing $\hat{\delta}_i$ by intervals of 0.1 until a solution is determined to be between $\hat{\delta}_i$ and $\hat{\delta}_{i-1}$. At this time a binary search for the solution to the appropriate equation is undertaken. In order to control computer time cost, an effort is made to find solutions correct to 10^{-5} within 50 iterations. Having found an estimate for δ , estimates for γ and β are easily calculated from equations (4.2) and (4.3).

For MLE and MMLE I-V, estimates of δ using equations (2.4), (3.4), (3.6), (3.7), (3.8), and (3.9) are calculated for specific values of $\hat{\gamma}$. The procedure used in searching for an estimate of δ for each specific value of $\hat{\gamma}$ is the same as that used in MOMENT. Having found an estimate of δ for a given $\hat{\gamma}$, equation (2.5) is employed to calculate $\hat{\beta}$. The search for estimates $\hat{\delta}$, $\hat{\gamma}$, and $\hat{\beta}$ begins with $\hat{\gamma} = x_1 - 10^{-4}$ and continues with $\hat{\gamma}$ being decremented until we find two values γ_i and γ_j such that $(\partial \ln L / \partial \gamma)_i \leq 0 \leq (\partial \ln L / \partial \gamma)_j$. Having found an upper and lower limit on $\hat{\gamma}$, a binary search is employed to calculate final estimates of γ which in turn are employed to calculate final estimates for δ and β .

8. A MONTE CARLO STUDY

A Monte Carlo study was undertaken in order to compare estimators with respect to their sampling behavior. One hundred random samples consisting of one hundred observations each were generated from Weibull populations for which $\delta = 4.5, 2.2156, 1.5639, 1.2111, 1.0$, and 0.8632 . (These values correspond to $\alpha_3 = -0.1784, 0.5, 1.0, 1.5, 2.0$ and 2.5 , respectively.) In addition 200 random samples consisting of 25 observations each were generated from populations in which $\delta = 1.0$ and 0.8632 . Ten samples of ten observations each were generated from a popu-

lation in which $\delta = 0.5$. The terminus γ was set equal to 0, and the scale parameter β was set equal to 1, but in calculating estimates all three parameters were assumed to be unknown. Note that four of the populations sampled are bell-shaped ($\delta > 1$) while three are reverse J-shaped ($\delta \leq 1$).

The pseudo-random number generator RANF designed by CDC for the 60-bit word Cyber 70/74 computer was used in the simulation procedure to generate uniformly distributed random numbers. The multiplier for this generator passes the Coveyou-Macpherson test and other statistical tests for randomness including the autocorrelation test with lag ≤ 100 and the triplet test. Details concerning these tests are given in Fortran Version 5 Common Library Mathematical Routine Reference Manual (1981), Control Data Corp. Minneapolis, Mn. CDC Operating Systems: Nos. 1, Nos./BC1, Scope 2, #60483100.

The uniformly distributed random numbers thus generated were subsequently transformed to produce Weibull distributed random observations from each of the populations sampled.

The software package described in Section 7 was employed in an effort to calculate applicable estimates from each sample generated. Because of the previously described restrictions in the computer programs, some samples failed to produce estimates from some estimators. The means and standard deviations of estimates from each sequence of samples, together with counts of samples for which no estimates were found are entered in Tables II and III. All computations were performed at the University of Georgia Computer Center on a Cyber 70/74 computer.

In order to compare the relative ease of computation for the various estimators under consideration, the average CPU per sample in seconds was recorded for five samples of size 100 from the population in which $\delta = 1.56391$. These are as follows:

ME	-	.180	MMLE-I	-	10.871
MME-I	-	.179	MMLE-II	-	12.322
MME-II	-	.182	MMLE-III	-	7.761
MME-III	-	.181	MMLE-IV	-	12.772
MLE	-	29.230	MMLE-V	-	17.181

TABLE IV
Characteristics of Populations Sampled
 $\gamma = 0, \beta = 1$

δ	μ_x	σ_x	Me:x	α_3
0.5000	2.0000	4.4721	0.4805	6.6188
0.8632	1.0777	1.2529	0.6540	2.5000
1.0000	1.0000	1.0000	0.6932	2.0000
1.2111	0.9385	0.7785	0.7389	1.5000
1.5639	0.8986	0.5871	0.7911	1.0000
2.2156	0.8856	0.4223	0.8475	0.5000
4.5000	0.9126	0.2291	0.9218	-0.1784

TABLE V
Random Sample From a Weibull Distribution
 $\gamma = 0, \beta = 1, \delta = 1.5639$

1.5102	1.7300	.2724	1.0941
.4386	1.1273	1.3218	.8744
.4081	.1257	.7082	.2159
.7649	1.6378	.6412	1.0712
.5281	2.7173	.4157	1.2513
1.0634	1.7553	1.2008	1.2513
1.4176	.9209	.3690	.7680
2.1820	1.0258	.3809	1.3782
.6674	.1009	1.0501	.5327
.6942	.6404	1.6469	.5520
.8721	.4124	.3857	.8587
2.4596	.7950	.8949	1.3545
.9890	1.5888	.5006	.2827
.7589	1.6565	.9640	.5513
2.3628	1.1701	1.1651	2.5450
.4272	.1854	.6158	.2795
.4869	.3329	.8367	1.0735
.1461	.5719	.2908	1.8398
.4874	.6130	.6993	.6460
1.5160	.7077	1.0976	.4527
.7245	1.1605	.9569	2.6288
.8242	1.0593	.2980	1.2859
.8230	1.6695	1.0899	1.5082
.4573	2.2271	.4852	.8858
1.7600	.9194	.5685	.3604

$n = 100, \bar{x} = 0.9602, s = 0.5961, a_3 = 1.0040, x_1 = 0.1009,$
 $x_{me} = 0.8477.$

Even though these times are averages of only five samples and are only for $\delta = 1.56391$ ($\alpha_3 = 1.0000$), indications based on total computer time for the various simulation runs involved in compiling Tables II and III suggest that these times are representative for the various estimators under study in this article.

A summary of the characteristics of the various populations sampled for this study is contained in Table IV which follows.

9. AN ILLUSTRATIVE EXAMPLE

The practical application of estimators presented in this paper is illustrated with a random sample consisting of 100 observations selected from a Weibull population in which $\gamma = 0$, $\beta = 1.0$, and $\delta = 1.5639$. The population mean, standard deviation, third standard moment, and median are 0.8986, 0.5871, 1.0, and 0.7911 respectively. Individual sample observations are listed in Table V. For this sample, $n = 100$, $\bar{x} = 0.9602$, $s = 0.5961$, $\alpha_3 = 1.0040$, $me_x = 0.8477$, and $x_1 = 0.1009$. The resulting estimates are displayed in Table VI. It is noted that for this sample, MME-I and MME-II yield estimates of δ that are

TABLE VI
Summary of Estimates

ESTIMATOR	γ	β	δ	μ_x	σ_x	$\alpha_3 : x$	Me_x
MLE	.084	0.970	1.501	.960	.594	1.070	.844
MMLE-I	.042	1.028	1.615	.963	.584	0.946	.861
MMLE-II	.050	1.017	1.595	.962	.586	0.968	.858
MMLE-III	.081	0.976	1.514	.961	.592	1.056	.847
MMLE-IV	.089	0.963	1.486	.959	.596	1.089	.841
MMLE-V	.078	0.980	1.523	.961	.591	1.046	.848
ME	.050	1.013	1.560	.960	.596	1.004	.851
MME-I	.047	1.016	1.565	.960	.596	0.999	.851
MME-II	.047	1.016	1.565	.960	.596	0.999	.851
MME-III	.064	0.995	1.534	.960	.596	1.033	.848
POPULATION	0	1	1.5639	.8986	.5871	1.000	.7911

in closest agreement with corresponding population values, while MME-III provides the closest estimate to the population value of β and MMLE-I provides the estimate of γ that is in closest agreement with the corresponding population value. However, the MME-I and MME-II estimates of γ are only slightly less favorable than the estimate obtained from MMLE-I.

10. CONCLUDING REMARKS AND RECOMMENDATIONS

An examination of the simulation summaries in Tables II and III discloses that when $\delta \geq 2.2156$ ($\alpha_3 < 0.5$), the MLE perform well with respect to bias and variance. For $\delta = 4.5$, both the ME and MLE perform quite well in these respects. However when computing time is considered, we note that the MLE required considerably more time than the ME. When $\delta \leq 2.2156$, MME-I, MMLE-I and MMLE-II appear to be the preferred estimators.

When computing time is not at a premium, it is recommended that MMLE-I or MMLE-II be employed regardless of the value of δ . If computing time is expensive, then it is recommended that MME-I be employed except in samples where skewness is near zero. In the latter situation the ME would be preferred over the modified estimators.

Both the ME and MME-I are easy to calculate on a simple hand-held calculator provided adequate tables of the gamma function are available. If a programmable hand-held calculator is available, the gamma function can be programmed and tables are not needed.

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